

HW0428

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10.18

a.

```
> cat("母親受過部分大學教育的比例：", mean(mroz1$MOTHERCOLL) * 100, "%\n")
母親受過部分大學教育的比例： 12.14953 %
> cat("父親受過部分大學教育的比例：", mean(mroz1$FATHERCOLL) * 100, "%\n")
父親受過部分大學教育的比例： 11.68224 %
```

b.

	educ	MOTHERCOLL	FATHERCOLL
educ	1.0000000	0.3594705	0.3984962
MOTHERCOLL	0.3594705	1.0000000	0.3545709
FATHERCOLL	0.3984962	0.3545709	1.0000000

因為在教育歷程，其是否就讀大學比起多讀一年書而言，較具鑑別度且具影響力。

c.

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
      exper + I(exper^2), data = mroz1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.08719	-0.32444	0.04147	0.36634	2.35621

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.1327561	0.4965325	-0.267	0.78932
educ	0.0760180	0.0394077	1.929	0.05440 .
exper	0.0433444	0.0134135	3.231	0.00133 **
I(exper^2)	-0.0008711	0.0004017	-2.169	0.03066 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6703 on 424 degrees of freedom
Multiple R-Squared: 0.147, Adjusted R-squared: 0.1409
Wald test: 8.2 on 3 and 424 DF, p-value: 2.569e-05

```
> confint(iv_model_c, "educ", level = 0.95)
      2.5 %      97.5 %
educ -0.001219763 0.1532557
```

雖然係數為正，但在 5%顯著水準為不顯著。

d.

```
Call:
lm(formula = educ ~ exper + I(exper^2) + MOTHERCOLL, data = mroz1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-7.4267	-0.4826	-0.3731	1.0000	4.9353

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.079094	0.303118	39.849	< 2e-16 ***
exper	0.056230	0.042101	1.336	0.182
I(exper^2)	-0.001956	0.001256	-1.557	0.120
MOTHERCOLL	2.517068	0.315713	7.973	1.46e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.133 on 424 degrees of freedom
Multiple R-squared: 0.1347, Adjusted R-squared: 0.1285
F-statistic: 21.99 on 3 and 424 DF, p-value: 2.965e-13

```
> # 執行 F-test 檢定 MOTHERCOLL 的係數是否為 0
> linearHypothesis(first_stage_d, "MOTHERCOLL = 0")
```

Linear hypothesis test:

MOTHERCOLL = 0

Model 1: restricted model

Model 2: educ ~ exper + I(exper^2) + MOTHERCOLL

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	424	1929.9	1	289.32	63.563	1.455e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

F 值遠大於其臨界值，MOTHERCOLL 是一個強工具變數，與教育高度相關。

c.

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
      I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.07797	-0.32128	0.03418	0.37648	2.36183

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.2790819	0.3922213	-0.712	0.47714
educ	0.0878477	0.0307808	2.854	0.00453 **
exper	0.0426761	0.0132950	3.210	0.00143 **
I(exper^2)	-0.0008486	0.0003976	-2.135	0.03337 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom

Multiple R-Squared: 0.153, Adjusted R-squared: 0.147

Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06

```
> confint(iv_model_e, "educ", level = 0.95)
```

	2.5 %	97.5 %
educ	0.02751845	0.1481769

此區間比 c 窄，表示增加有效工具變數能提高估計準確度，並使教育對薪資的正向影響在 5% 水準下變顯著 ($p = 0.00453$)。

f.

```
Call:
lm(formula = educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL,
    data = mroz1)

Residuals:
    Min       1Q   Median       3Q      Max
-7.2152 -0.3056 -0.2152  0.7627  5.0620

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 11.890259   0.290251  40.965 < 2e-16 ***
exper        0.049149   0.040133   1.225  0.221
I(exper^2)  -0.001449   0.001199  -1.209  0.227
MOTHERCOLL   1.749947   0.322347   5.429 9.58e-08 ***
FATHERCOLL   2.186612   0.329917   6.628 1.04e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.033 on 423 degrees of freedom
Multiple R-squared:  0.2161,    Adjusted R-squared:  0.2086
F-statistic: 29.15 on 4 and 423 DF,  p-value: < 2.2e-16

> # 檢定 H0: MOTHERCOLL = 0 且 FATHERCOLL = 0
> linearHypothesis(first_stage_f, c("MOTHERCOLL = 0", "FATHERCOLL = 0"))

Linear hypothesis test:
MOTHERCOLL = 0
FATHERCOLL = 0

Model 1: restricted model
Model 2: educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL

   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     425 2219.2
2     423 1748.3  2     470.88 56.963 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F 值為 56.963 遠大於其臨界值，依然非常強勁。

g.

Call:

```
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +  
      I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.07797	-0.32128	0.03418	0.37648	2.36183

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.2790819	0.3922213	-0.712	0.47714
educ	0.0878477	0.0307808	2.854	0.00453 **
exper	0.0426761	0.0132950	3.210	0.00143 **
I(exper^2)	-0.0008486	0.0003976	-2.135	0.03337 *

Diagnostic tests:

	df1	df2	statistic	p-value
Weak instruments	2	423	56.963	<2e-16 ***
Wu-Hausman	1	423	0.519	0.472
Sargan	1	NA	0.238	0.626

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom

Multiple R-Squared: 0.153, Adjusted R-squared: 0.147

Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06

虛無假設：Mothercoll 與 Fathercoll 皆為有效工具變數（意即工具變數與誤差項不相關）

Sargan Test 的 P value (0.626) 大於 0.05，無法拒絕虛無假設。代表這兩個工具變數是有效的，因此模型設定算良好。

10.20

a.

```
Call:
lm(formula = msft_riskpremium ~ mkt_riskpremium, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27424 -0.04744 -0.00820  0.03869  0.35801

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.003250   0.006036   0.538   0.591
mkt_riskpremium 1.201840   0.122152   9.839 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom
Multiple R-squared:  0.3523,    Adjusted R-squared:  0.3486
F-statistic: 96.8 on 1 and 178 DF,  p-value: < 2.2e-16
```

因為 $\beta = 1.201840 > 1$ ，微軟相對市場投資組合而言為風險性(Risky)股票。

b.

```
Call:
lm(formula = mkt_riskpremium ~ RANK, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.110497 -0.006308  0.001497  0.009433  0.029513

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.903e-02  2.195e-03  -36.0 <2e-16 ***
RANK          9.067e-04  2.104e-05   43.1 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared:  0.9126,    Adjusted R-squared:  0.9121
F-statistic: 1858 on 1 and 178 DF,  p-value: < 2.2e-16
```

RANK 與市場風險溢酬高度相關，是一個極強的工具變數。其 R 平方為 0.9126

c.

```
Call:
lm(formula = msft_riskpremium ~ mkt_riskpremium + v_hat, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27140 -0.04213 -0.00911  0.03423  0.34887

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.003018   0.005984   0.504   0.6146
mkt_riskpremium 1.278318   0.126749  10.085 <2e-16 ***
v_hat        -0.874599   0.428626  -2.040   0.0428 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672,    Adjusted R-squared:  0.36
F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

在 1%顯著水準下，v_hat 是不顯著的，說明市場溢酬應被視是外生變數。

d.

```
Call:
ivreg(formula = msft_riskpremium ~ mkt_riskpremium | RANK, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.271625 -0.049675 -0.009693  0.037683  0.355579

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.003018   0.006044   0.499   0.618
mkt_riskpremium 1.278318   0.128011   9.986 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-Squared:  0.3508,    Adjusted R-squared: 0.3472
Wald test: 99.72 on 1 and 178 DF,  p-value: < 2.2e-16
```

此值高於 OLS 的 1.201840。這符合衰減偏誤(Attenuation Bias)的預期：當自變數有測量誤差時，OLS 會低估係數，而 IV 能將其修正回來。

c.

```
Call:
lm(formula = mkt_riskpremium ~ RANK + POS, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.109182 -0.006732  0.002858  0.008936  0.026652

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0804216   0.0022622  -35.55  <2e-16 ***
RANK         0.0009819   0.0000400   24.55  <2e-16 ***
POS        -0.0092762   0.0042156   -2.20   0.0291 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared:  0.9149,    Adjusted R-squared:  0.9139
F-statistic: 951.3 on 2 and 177 DF,  p-value: < 2.2e-16

> # 講解：檢定 RANK 與 POS 的聯合顯著性 (F-test)。
> linearHypothesis(first_stage_e, c("RANK=0", "POS=0"))

Linear hypothesis test:
RANK = 0
POS = 0

Model 1: restricted model
Model 2: mkt_riskpremium ~ RANK + POS

   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     179 0.43784
2     177 0.03727   2    0.40057 951.26 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

RANK 與 POS 為強大的工具變數組合。第一階段 R 平方微幅上升至 0.9149。

f.

```
Call:
lm(formula = msft_riskpremium ~ mkt_riskpremium + v_hat_e, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27132 -0.04261 -0.00812  0.03343  0.34867

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003004   0.005972   0.503   0.6157
mkt_riskpremium 1.283118   0.126344  10.156  <2e-16 ***
v_hat_e      -0.954918   0.433062  -2.205   0.0287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared:  0.3696,    Adjusted R-squared:  0.3625
F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16
```

在 1% 顯著水準下，市場報酬是外生的（因為殘差的 p-value 0.0287 > 0.01）。

g.

```
Call:
ivreg(formula = msft_riskpremium ~ mkt_riskpremium | RANK + POS,
      data = capm5)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.27168	-0.04960	-0.00983	0.03762	0.35543

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003004	0.006044	0.497	0.62
mkt_riskpremium	1.283118	0.127866	10.035	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08093 on 178 degrees of freedom
Multiple R-Squared: 0.3507, Adjusted R-squared: 0.347
Wald test: 100.7 on 1 and 178 DF, p-value: < 2.2e-16

估計值進一步增加。加強「OLS 因測量誤差而低估微軟風險」的觀點。IV 估計出的 Beta 更能反映微軟與市場真實的敏感度。

h.

```
Sargan Test Statistic: 0.5584634
> cat("P-value:", p_value_sargan, "\n")
P-value: 0.45488
```

$p\text{-value} = 0.45488 > 0.05$ ，則可認為工具變數具有有效性。